

Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering

09/2023 – Present

Kabataş Boys' High School

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00.
- Actively participated in various extracurricular activities and student societies, including Debate, Model United Nations, Entrepreneurship, and the Coding Club.

Experience

Executive Board Member at Radyo Boğaziçi

09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

Work and Travel Program Participant

06/2025 – 09/2025

- Gained valuable international experience through the cultural exchange and work program in the USA which improved my communication, English, teamwork and problem-solving skills.

Regional Organizational Team Member at ESTIEM

09/2023 – Present

- Contributed to the planning and coordination of events, team communication, and organizational processes within ESTIEM, an international network connecting Industrial Engineering and Management students across Europe.

Technical Projects

AlphaTrigger AI Terminal — Python, FastAPI, Next.js, WebSockets

- Autonomous high-frequency trading simulation engine featuring a deterministic V8 Risk-First strategy, dynamic volatility-adjusted position sizing, and a real-time WebSocket-driven dashboard.

Performo Enterprise Admin — Next.js 15+, Prisma, Tailwind CSS, Framer Motion

- Engineered a high-density administrative intelligence dashboard with real-time operational metrics and server-side synchronized state management for high-intensity monitoring.
- Implemented a professional "Triple Elite" design system with sub-second data refresh cycles via force-dynamic rendering and optimized SQL indexing for administrative stability.

AI-Powered FinTech Intelligence Platform — Next.js, FastAPI, PostgreSQL, Docker

- Architected a full-stack, deployment-ready financial dashboard combining personal finance analytics, an algorithmic portfolio risk evaluator, and OpenAI-generated insights with secure NLP fallbacks.
- Engineered a live limit-order trading simulator featuring a synchronous Price-Time Priority matching engine and achieved 100% backend test coverage with automated CI/CD pipelines.

SchedSim CPU Scheduler — x86-64 Assembly, GNU Toolchain, Linux Syscalls

- High-performance CPU scheduling simulation engine (FCFS, SJF, SRTF, PF, RR) built with zero standard library dependencies, utilizing direct Linux syscalls for I/O and process management.

Spirelayer Command Interpreter — C, Custom Backtracking Lexer

- Robust CLI-based interpreter replicating fantasy RPG REPL mechanics; implemented an autonomous backtracking lexical parser to handle non-deterministic grammar and complex command resolution.

Aura Functional Interpreter — Python, Functional Paradigms, Dual Scoping Engine

- Advanced interpreter for a custom functional language supporting first-class closures, higher-order function patterns, and toggleable lexical/dynamic scoping engines for complex variable resolution.

MatrixNet — Java, Tarjan's SCC Algorithm, Graph Theory

- Network graph analysis tool implementing Tarjan's SCC and Articulation Point detection ($O(V + E)$); optimized with Open Addressing to eliminate internal JVM string allocation overheads.

GigMatch-Pro — Java, Memory-Optimized Heaps

- Scalable task-matching marketplace engine utilizing optimized dynamic Max-Heaps for high-efficiency priority allocation and candidate-to-job matching algorithms.

Nightpass Survival Game — Java, Rule Engine Architecture

- Algorithmic implementation of a complex rule engine governing strategic deck, hand, and resource economics in a terminal-based survival card game environment.

Volunteer Experience

Education Mentor at TOÇEV 09/2019 – 03/2020

- Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds.

Volunteer at Kabataş Boys’ High School Community Involvement Projects Club 09/2018 – 06/2023

- Carried out various volunteer projects aimed at helping and supporting people in need through community involvement initiatives.

Volunteering Division Member at Boğaziçi University Industrial Engineering Club 09/2023 – 06/2024

- Supported the organization and execution of volunteer initiatives and community-oriented projects within the club.

Education Mentor at Derece Kampüsü 09/2023 – 09/2025

- Mentored high school students for the university entrance exams, providing academic guidance and exam strategy support.

Technical Skills

- **Low-Level & Systems:** x86-64 Assembly, C, Linux Syscalls, Process Scheduling, Multithreading (Pthreads), Memory Management
- **Enterprise Full-Stack:** Next.js 15+, TypeScript, React, Prisma (ORM), SQLAlchemy (ORM), FastAPI, Tailwind CSS, Framer Motion, PostgreSQL, SQLite
- **Data Structures & Algos:** Tarjan’s SCC, Articulation Points, Max-Heaps, Complexity Analysis (Big O), Graph Theory, Recursive Optimization
- **Paradigms & Models:** Functional (Scheme/Racket), Logic (Prolog), OOP (Java/C++), Interpreter Design, State Machines
- **AI & Modern Engineering:** OpenAI API Integration, Advanced Agentic Coding, LLM Prompting, AI-Assisted Architecture, Autonomous Feature Deployment
- **Tools & Ecosystem:** Linux/Unix, Git/GitHub, Docker, Bash, Alembic, Pytest (Testing), CI/CD, Makefile, LaTeX, Adobe Photoshop

Languages

- Turkish (native)
- English (fluent)
- German (elementary)

Extracurricular Activities

- DJ, Radyo Boğaziçi
- Student-Athlete, Boğaziçi University Sailing Team
- Student-Athlete, Boğaziçi University Ski Team
- Windsurf Athlete, Competing Individually